**King Fahd University of Petroleum and Minerals**
**Department of Mathematics**
*Student effort not Official. Generated by Ice ❄. دعواتكم له*

# Math 208 Major Exam II Term 233

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

---

**1.** The rank of the matrix $A = \begin{bmatrix} 1 & 2 & 1 & 3 & 2 \\ 3 & 4 & 9 & 0 & 7 \\ 2 & 3 & 5 & 1 & 8 \\ 2 & 2 & 8 & -3 & 5 \end{bmatrix}$ is

(a) 5

(b) 1

(c) 4

(d) 2

(e) 3

**2.** If the characteristic polynomial of the matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 7 & 2 \\ 10 & -15 & -4 \end{bmatrix}$$

is $p(\lambda) = -(\lambda - 1)^2(\lambda - 2)$, then a basis for the eigenspace of $\lambda = 1$ is

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ \alpha \end{bmatrix}, \; v_2 = \begin{bmatrix} \beta \\ 2 \\ 0 \end{bmatrix},$$

then $\alpha + \beta =$

(a) 5

(b) 0

(c) 4

(d) $-4$

(e) $-5$

**3.** The general solution of the differential equation

$$2y'' - y' - y = 0$$

is

(a) $y(x) = c_1 e^{-x/2} + c_2 e^{-x}$

(b) $y(x) = c_1 e^{x/2} + c_2 e^{x}$

(c) $y(x) = c_1 e^{x} + c_2 x e^{x}$

(d) $y(x) = c_1 e^{-x/2} + c_2 e^{x}$

(e) $y(x) = c_1 e^{x} + c_2 e^{-x}$

**4.** The solution of the initial-value problem

$$y'' - 6y' + 25y = 0; \quad y(0) = 3,\ y'(0) = 1$$

is

(a) $y(x) = e^{3x}[3\cos(4x) + 2\sin(4x)]$

(b) $y(x) = e^{3x}[3\cos(4x) - 2\sin(4x)]$

(c) $y(x) = e^{-2x}[3\cos(4x) - 2\sin(4x)]$

(d) $y(x) = e^{-3x}[3\cos(4x) - 2\sin(4x)]$

(e) $y(x) = e^{2x}[3\cos(4x) - 2\sin(4x)]$

**5.** The characteristic polynomial of the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 7 & 2 \\ 10 & -15 & -4 \end{bmatrix}$ is $P(\lambda) =$

(a) $-\lambda^3 + 4\lambda^2 + 5\lambda - 2$

(b) $\lambda^3 + 4\lambda^2 + 5\lambda + 2$

(c) $-\lambda^3 - 4\lambda^2 - 5\lambda + 2$

(d) $-\lambda^3 + 4\lambda^2 - 5\lambda + 2$

(e) $\lambda^3 + 4\lambda^2 - 5\lambda + 2$

**6.** Consider the subspace $S$ of $\mathbb{R}^4$ defined by $S = \{(a, b, c, d) \mid a = 3c,\ b = 4d\}$. A basis of $S$ consists of the vectors

(a) $v_1 = (3, 0, -1, 0),\ v_2 = (0, -4, 0, 1)$

(b) $v_1 = (3, 0, 1, 0),\ v_2 = (0, 4, 0, 1)$

(c) $v_1 = (3, 0, 1, 0),\ v_2 = (0, 4, 0, -1)$

(d) $v_1 = (4, 0, 1, 0),\ v_2 = (0, 4, 0, 1)$

(e) $v_1 = (-3, 0, 1, 0),\ v_2 = (0, 4, 0, 1)$

**7.** The general solution of the differential equation $y^{(4)} - 8y'' + 16y = 0$ is

(a) $y(x) = c_1e^{2x} + c_2xe^{-2x} + c_3xe^{-2x} + c_4e^x$

(b) $y(x) = (c_1 + c_2 + c_3x^2 + c_4x^3)e^{2x}$

(c) $y(x) = c_1e^{2x} + c_2xe^{2x} + c_3e^{-2x} + c_4xe^{-2x}$

(d) $y(x) = c_1e^{2x} + c_2xe^{2x} + c_3e^{-2x} + c_4e^{-x}$

(e) $y(x) = c_1e^{2x} + c_2xe^{2x} + c_3e^{-2x} + c_4e^x$

**8.** If the solution space of the system

$$\begin{aligned} x_1 - 3x_2 - 10x_3 + 5x_4 &= 0 \\ x_1 + 4x_2 + 11x_3 - 2x_4 &= 0 \\ x_1 + 3x_2 + 8x_3 - x_4 &= 0 \end{aligned}$$

has all linear combination of the two vectors $u = (1, \alpha, 1, 0)$ and $v = (-2, \beta, 0, 1)$, then $\alpha + \beta =$

(a) $-4$

(b) $-2$

(c) 0

(d) 4

(e) 2

**9.** The eigenvector associated with eigenvalue $\lambda = 1$ of the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ -6 & 9 & 3 \\ 4 & -8 & -1 \end{bmatrix}$ is $\begin{bmatrix} 1 \\ 0 \\ \alpha \end{bmatrix}$, where $\alpha =$

(a) 1

(b) $-2$

(c) $-3$

(d) 2

(e) 3

**10.** If $W(x)$ is the wronskian of the functions

$$f(x) = e^x,\ g(x) = e^{-x},\ h(x) = e^{2x},$$

then $W(0) =$

(a) 8

(b) $-6$

(c) $-4$

(d) 4

(e) 6

**11.** Given that $y_p = u_1(x)\cos x + u_2(x)\sin x$ is a particular solution of the differential equation $y'' + y = \csc^2 x$, then $u_2(x) =$

(a) $\sec x$

(b) $-\sin x$

(c) $-\csc x \cot x$

(d) $\cos x$

(e) $-\csc x$

**12.** An appropriate form of a particular solution $y_p$ for the non-homogeneous differential equation

$$(D-1)^3(D^2-4)\,y = xe^x + e^{2x} + e^{-2x}$$

is given by $y_p(x) =$

(a) $Ax^3e^x + Bx^4e^x + Ce^{2x} + Dxe^{-2x}$

(b) $Ax^3e^x + Bx^4e^x + Cxe^{2x} + De^{-2x}$

(c) $Ax^3e^x + Bx^4e^x + Cxe^{2x} + Dxe^{-2x}$

(d) $Axe^x + Bx^2e^x + Cxe^{2x} + Dxe^{-2x}$

(e) $Ax^2e^x + Bx^3e^x + Cxe^{2x} + Dxe^{-2x}$

**13.** If the matrix $A = \begin{bmatrix} 10 & -6 \\ 12 & -7 \end{bmatrix}$ is diagonalizable with a diagonalizing matrix $P$ and a diagonal matrix $D$ such that $P^{-1}AP = D$, then

(a) $P = \begin{bmatrix} 2 & 3 \\ 3 & 3 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$

(b) $P = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(c) $P = \begin{bmatrix} 3 & 3 \\ 3 & 4 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(d) $P = \begin{bmatrix} 2 & 3 \\ 3 & 3 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(e) $P = \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

**14.** A linear homogeneous constant-coefficient differential equation which has the general solution

$$y(x) = A\cos(2x) + B\sin(2x) + C\cosh(2x) + D\sinh(2x)$$

is

(a) $y^{(4)} - 64y = 0$

(b) $y^{(4)} - 16y = 0$

(c) $y^{(4)} + 7y'' - 144y = 0$

(d) $y^{(4)} - 7y'' + 144y = 0$

(e) $y^{(4)} - 5y^{(3)} + 4y'' - 3y' + y = 0$

**15.** If $y_p = A + Bx + Cx^2$ is a particular solution of the differential equation

$$5y'' + 4y' + 3y = 2 + x^2,$$

then $27A + 9B + 3C =$

(a) 12

(b) $-14$

(c) $-12$

(d) 13

(e) 0

# Answer Key

*Student effort not Official. Generated by Ice ❄. دعواتكم له*

| Q | Ans | Brief explanation |
|---|---|---|
| 1 | **(e)** | Row reduction leaves 3 nonzero rows, so rank $A = 3$. |
| 2 | **(a)** | $A - I$ gives $2x - 3y - z = 0$; $\alpha = 2$, $\beta = 3$, sum 5. |
| 3 | **(d)** | $2r^2 - r - 1 = (2r + 1)(r - 1)$, roots $-\frac{1}{2}, 1$. |
| 4 | **(b)** | $r = 3 \pm 4i$; $c_1 = 3$, $3c_1 + 4c_2 = 1 \Rightarrow c_2 = -2$. |
| 5 | **(d)** | $\det(A - \lambda I) = -(\lambda - 1)^2(\lambda - 2) = -\lambda^3 + 4\lambda^2 - 5\lambda + 2$. |
| 6 | **(b)** | $(3c, 4d, c, d) = c(3, 0, 1, 0) + d(0, 4, 0, 1)$. |
| 7 | **(c)** | $(r^2 - 4)^2 = 0$: double roots $\pm 2$. |
| 8 | **(b)** | Free $x_3, x_4$: $u = (1, -3, 1, 0)$, $v = (-2, 1, 0, 1)$, so $\alpha + \beta = -2$. |
| 9 | **(d)** | $(A - I)(1, 0, \alpha)^T = 0$: $-6 + 3\alpha = 0$, so $\alpha = 2$. |
| 10 | **(b)** | $W(0) = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 1 & 1 & 4 \end{vmatrix} = -6.$ |
| 11 | **(e)** | $W = 1$, $u_2' = \frac{\cos x}{\sin^2 x}$, so $u_2 = -\csc x$. |
| 12 | **(c)** | Roots 1 (triple), 2, $-2$: multiply by $x^3$, $x$, $x$ respectively. |
| 13 | **(b)** | $\lambda = 1, 2$ with eigenvectors $(2, 3), (3, 4)$ as columns of $P$. |
| 14 | **(b)** | Roots $\pm 2i, \pm 2$: $(r^2 + 4)(r^2 - 4) = r^4 - 16$. |
| 15 | **(d)** | $C = \frac{1}{3}$, $B = -\frac{8}{9}$, $A = \frac{20}{27}$, so $27A + 9B + 3C = 13$. |